higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PLUMBING
NQF LEVEL 2

(12020162)

12 March 2019 (X-Paper)
09:00–12:00

Calculators and drawing instruments may be used.

This question paper consists of 6 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches should be neat, labelled and large enough so that ALL details are clear.
5. Start each question on a NEW page.
6. Write neatly and legibly.

NCV

QUESTION 1: INTRODUCTION TO PLUMBING

1.1 Explain the term *waste water*. (2)

1.2 Name the phases of the water cycle. (3)

1.3 What is the first process in the primary treatment of sewage? (1)

1.4 List FOUR roles of a plumbing project manager. (4)

1.5 Name FIVE relevant tradespeople with whom plumbers interact. (5)

[15]

QUESTION 2: PLUMBING PRINCIPLES AND CODE OF PRACTICE

2.1 Give THREE examples of surface water. (3)

2.2 Explain how water flows through a pipe. (2 × 2) (4)

2.3 Name the type of plant used to purify seawater. (1)

2.4 Name THREE systems that offer a certificate of approval. (3)

2.5 Which formula is used to calculate the pressure of water at a point? (2)

2.6 What is meant by the term *water-reticulation system*? (2)

[15]

QUESTION 3: MATERIALS, TOOLS AND EQUIPMENT

3.1 The sketches below show tools used in the plumbing industry.

Write only the name of each tool next to the question number (3.1.1–3.1.2) in the ANSWER BOOK.

3.1.1



3.1.2



(2 × 1) (2)

3.2 Differentiate between *capillary fittings* and *brass compressor fittings* (according to their classes). (2)

3.3 Give THREE examples of waste-water fixtures. (3)

3.4 Give TWO examples of soil-water fixtures. (2)

3.5 Name ONE type of scaffold. (1)
[10]

QUESTION 4: WATER-PIPE ASSEMBLY

4.1 What is the purpose of reaming the inside of a GMS pipe after threading it? (1)

4.2 Describe the procedure to follow when joining two copper pipes using a capillary fitting. (10)

4.3 A 90° bend is to be made in a copper pipe with a 22 mm diameter. The distance between the end of the pipe and the inside of the bend is 745 mm.

Calculate the distance at which the bending mark should be made on the pipe? (1)

4.4 Name TWO common methods to bend a multilayer pipe. (2)

4.5 Briefly explain what each of the following abbreviations stand for:

4.5.1 mPVC

4.5.2 uPVC

4.5.3 HDPE

(3 × 1) (3)

4.6 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (4.6.1–4.6.5) in the ANSWER BOOK.

4.6.1 Multilayer pipes can be cut with a sharp knife.

4.6.2 Multilayer pipes are suitable for use in underground drainage.

4.6.3 Multilayer pipes are made of copper.

4.6.4 Multilayer pipes can be bent.

4.6.5 Flexible monolayer pipes are suitable for use in hot water.

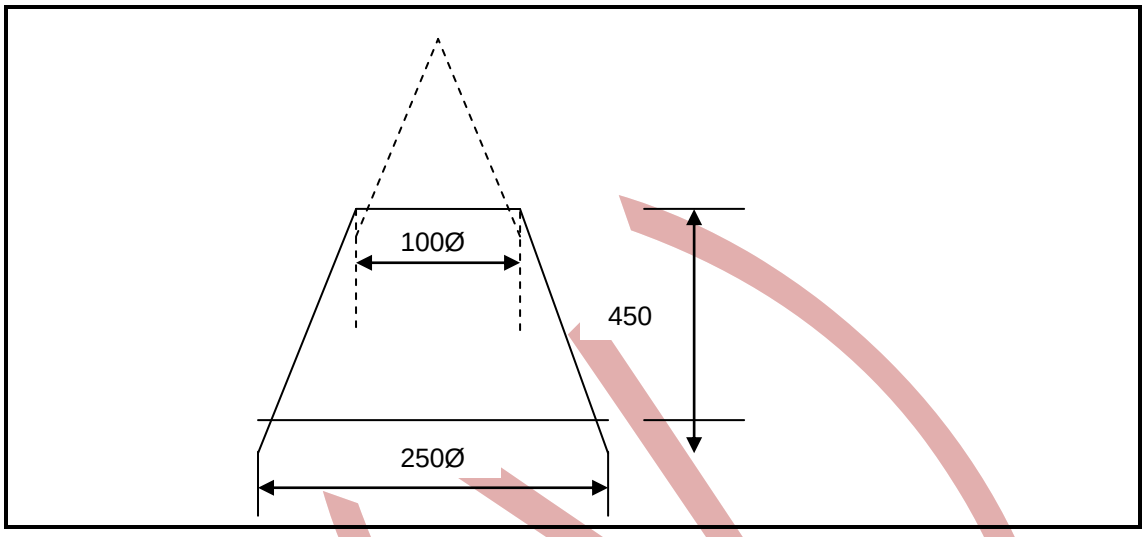
(5 × 1) (5)

4.7 State THREE ways of joining monolayer pipes. (3)
[25]

Copyright reserved

QUESTION 5: CUTTING AND JOINING OF METALS

5.1 Draw, to scale 1:5, the full pattern of the cone shown below in the ANSWER BOOK.



(8)

5.2 Name the TWO substances of which the flux used for brazing purposes consist. (2)

[10]

QUESTION 6: SET OUT BELOW-GROUND DRAINAGE SYSTEMS

6.1 Give a reason why access points are made when drains are laid out. (2)

6.2 Which part of a gulley collects some of the grease and dirt in the water before throwing it into the underground drainage systems? (1)

6.3 Briefly explain each of the following terms:

6.3.1 Storm water

6.3.2 Foul water

(2 × 2) (4)

6.4 List SIX main points to consider when laying out a drainage system. (6)

6.5 What is the maximum depth of an inspection chamber? (1)

6.6 Draw line diagrams of the symbols for each of the following terms:

6.6.1 Centre line

6.6.2 Diameter (inside)

6.6.3 Diameter (outside)

(3 × 1) (3)

Copyright reserved

6.7 List THREE items needed to create a water level to determine levels on a building site. (3)

6.8 Give the formula used to calculate the fall of a drain. (1)

6.9 What is the function of the bedding of a drainpipe? (2)

6.10 Briefly explain each of the following terms:

6.10.1 Invert level

6.10.2 Crown

(2 × 1) (2)
[25]

TOTAL: 100

NC(V)

Copyright reserved